

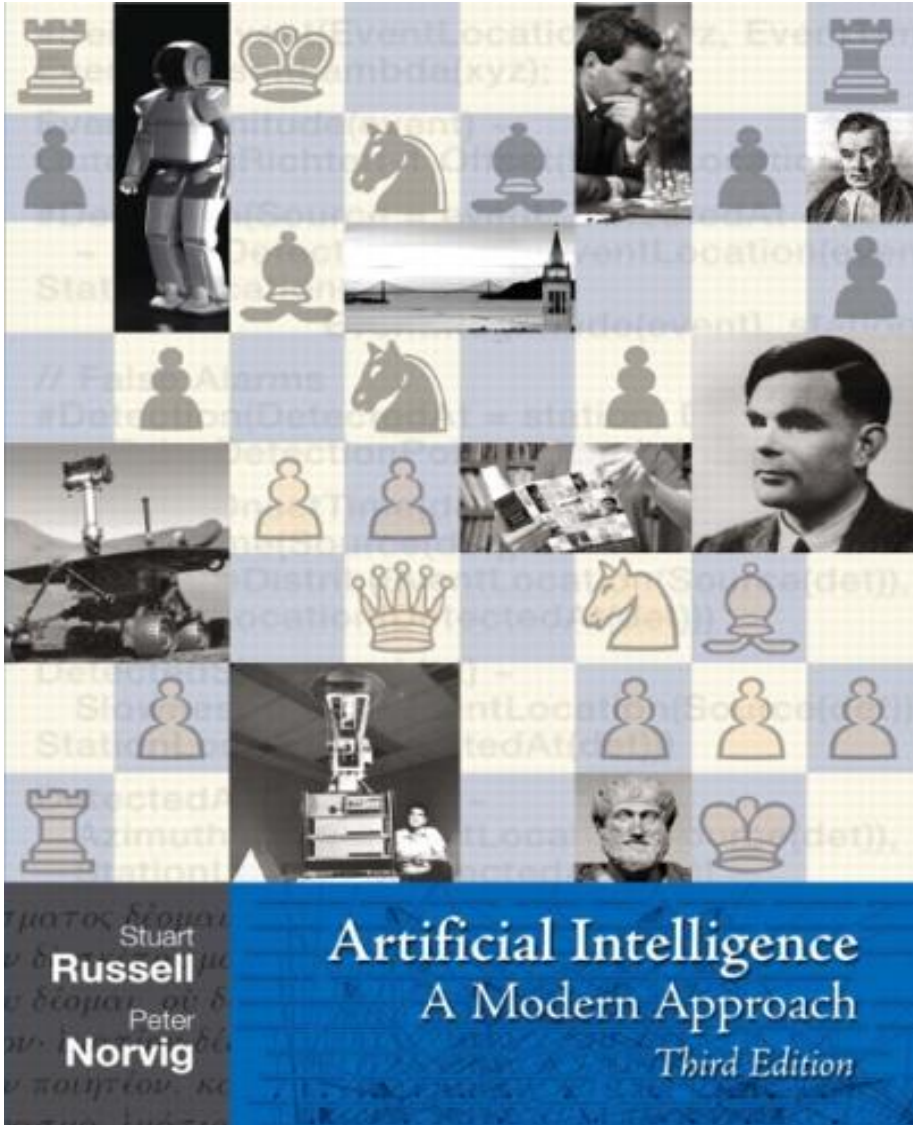
Artificial Intelligence



Chapter 1

Introduction

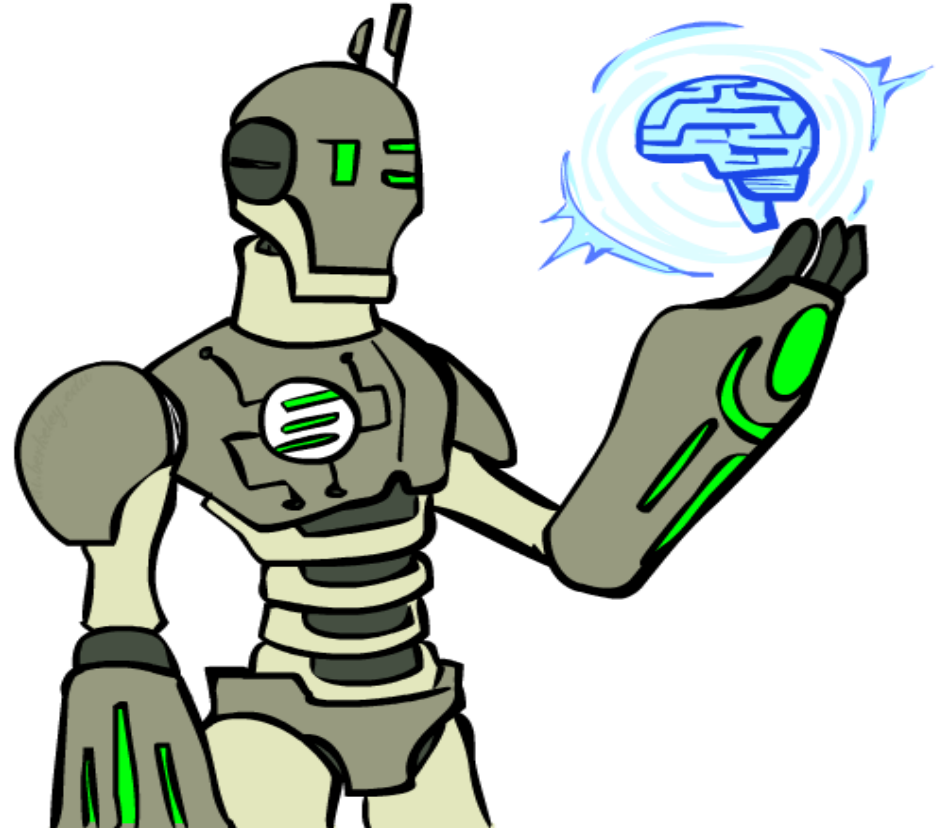
Artificial Intelligence



Russell & Norvig, AI: A Modern Approach, 3rd Ed.

Artificial Intelligence

- What is artificial intelligence?
- What can AI do?

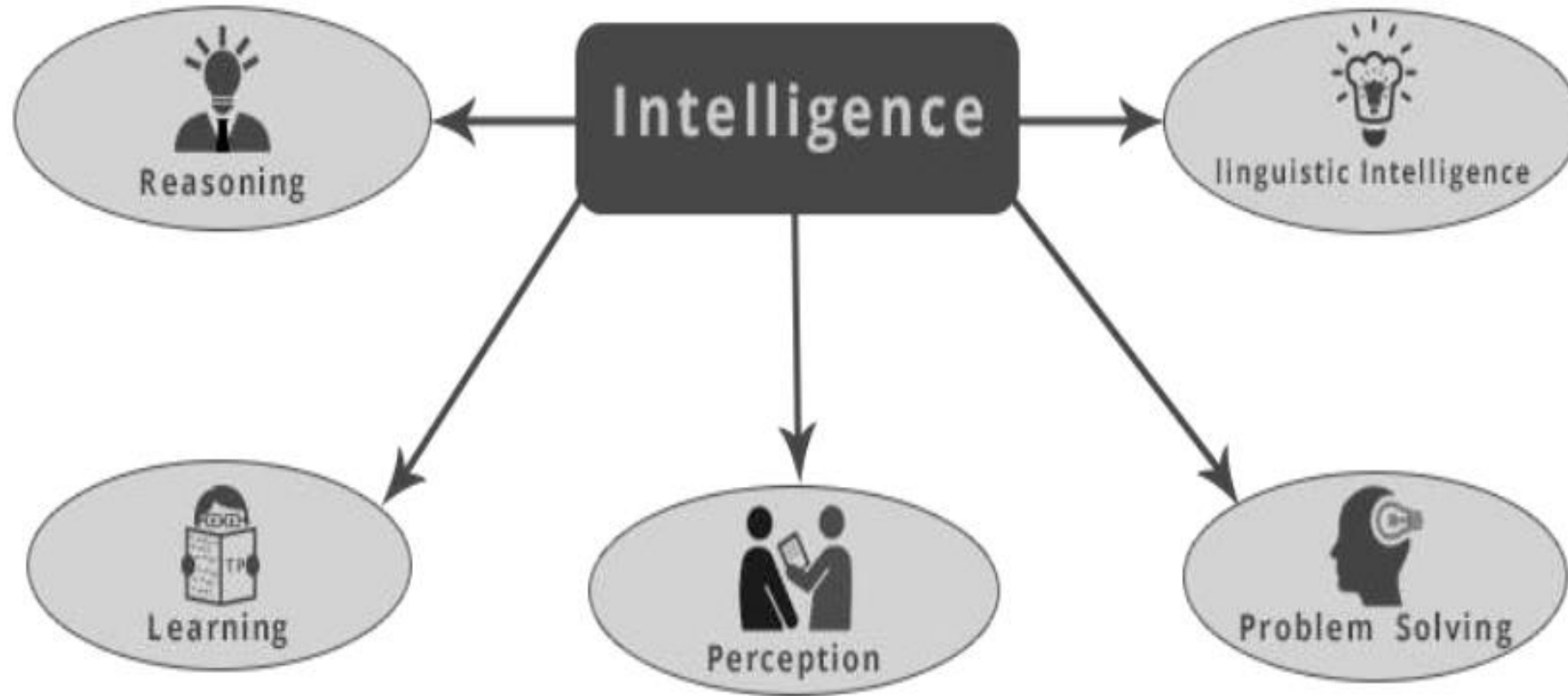


INTELLIGENCE

- We call ourselves *Homo sapiens*—man the wise—because our **intelligence** is so important to us.
- For thousands of years, we have tried to understand *how we think*; that is, how a mere handful of matter can perceive, understand, predict, and manipulate a world far larger and more complicated than itself.



INTELLIGENCE



The field of **artificial intelligence** (AI)

- Attempts not just to understand but also to *build* intelligent entities.
- AI is one of the newest fields in science and engineering. Work started in earnest soon after World War II, and the name itself was coined in 1956. Along with molecular biology, AI is regularly cited as the “field I would most like to be in” by scientists in other disciplines.
- AI currently encompasses a huge variety of subfields, ranging from the general (learning and perception) to the specific, such as playing chess, proving mathematical theorems, writing poetry, driving a car on a crowded street, and diagnosing diseases. AI is relevant to any intellectual task; it is truly a universal field.



Artificial Intelligence

Deep Learning

Natural Language Processing

Machine Learning

Neural Networks

Computer Vision



Supervised Learning



Unsupervised Learning

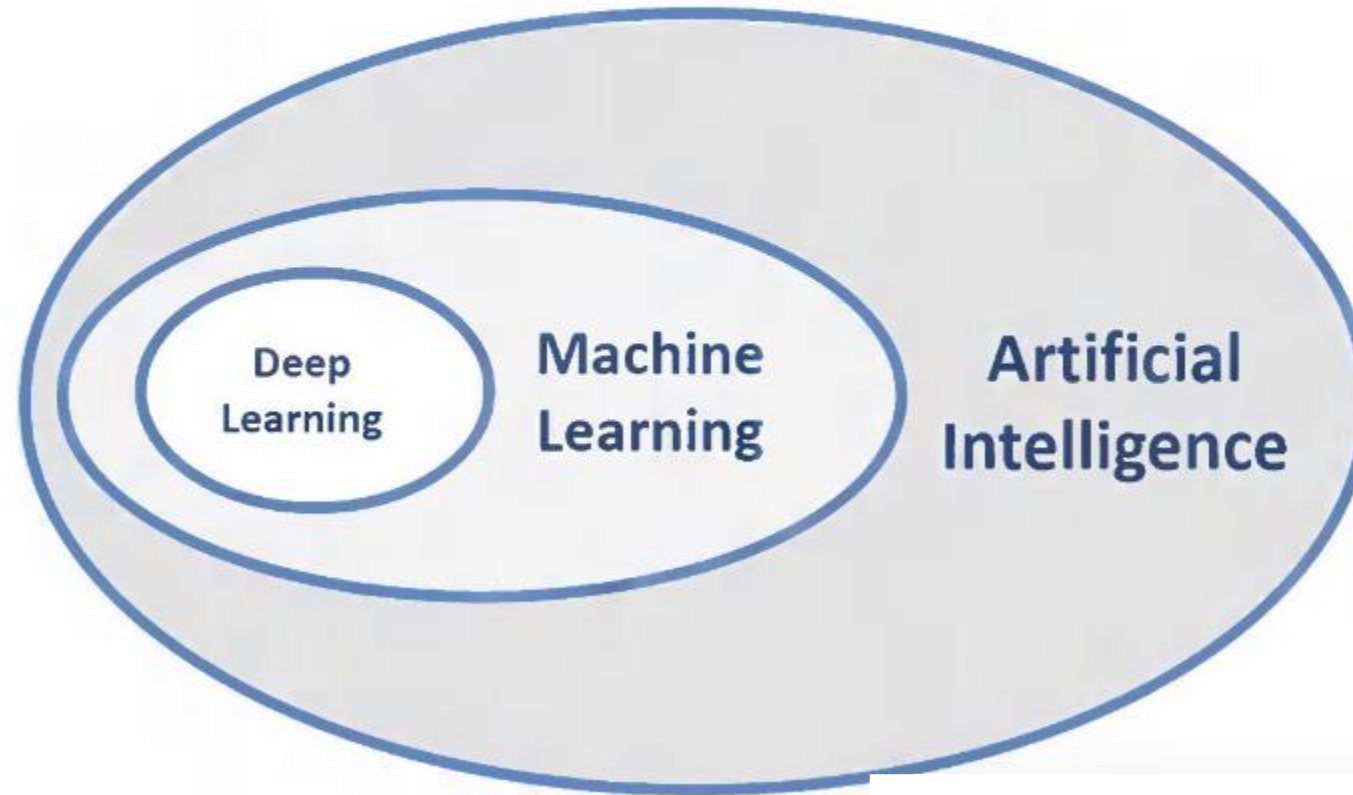


Reinforcement Learning



Semi-supervised Learning

What is the relationship between AI, ML, and DL?



Histroy

Artificial Intelligent Time Line

<https://verloop.io/blog/the-timeline-of-artificial-intelligence-from-the-1940s/>

Types of Artificial Intelligence

3 Types of Artificial Intelligence

Artificial Narrow Intelligence (ANI)



Stage-1

Machine Learning

- Specialises in one area and solves one problem



Siri



Alexa



Cortana

Artificial General Intelligence (AGI)



Stage-2

Machine Intelligence

- Refers to a computer that is as smart as a human across the board

Artificial Super Intelligence (ASI)



Stage-3

Machine Consciousness

- An intellect that is much smarter than the best human brains in practically every field

There are three types of Artificial Intelligence

- Artificial Narrow Intelligence (ANI)
- Artificial General Intelligence (AGI)
- Artificial Super Intelligence (ASI)

What is Artificial Narrow Intelligence (ANI)?

This is the most common form of AI that you'd find in the market now. These Artificial Intelligence systems are designed to solve one single problem and would be able to execute a single task really well. By definition, they have narrow capabilities, like recommending a product for an e-commerce user or predicting the weather. This is the only kind of Artificial Intelligence that exists today. They're able to come close to human functioning in very specific contexts, and even surpass them in many instances, but only excelling in very controlled environments with a limited set of parameters.

What is Artificial General Intelligence (AGI)?

AGI is still a theoretical concept. It's defined as AI which has a human-level of cognitive function, across a wide variety of domains such as language processing, image processing, computational functioning and reasoning and so on.

We're still a long way away from building an AGI system. An AGI system would need to comprise of thousands of Artificial Narrow Intelligence systems working in tandem, communicating with each other to mimic human reasoning. Even with the most advanced computing systems and infrastructures, such as Fujitsu's K or IBM's Watson, it has taken them 40 minutes to simulate a single second of neuronal activity. This speaks to both the immense complexity and interconnectedness of the human brain, and to the magnitude of the challenge of building an AGI with our current resources.

What is Artificial Super Intelligence (ASI)?

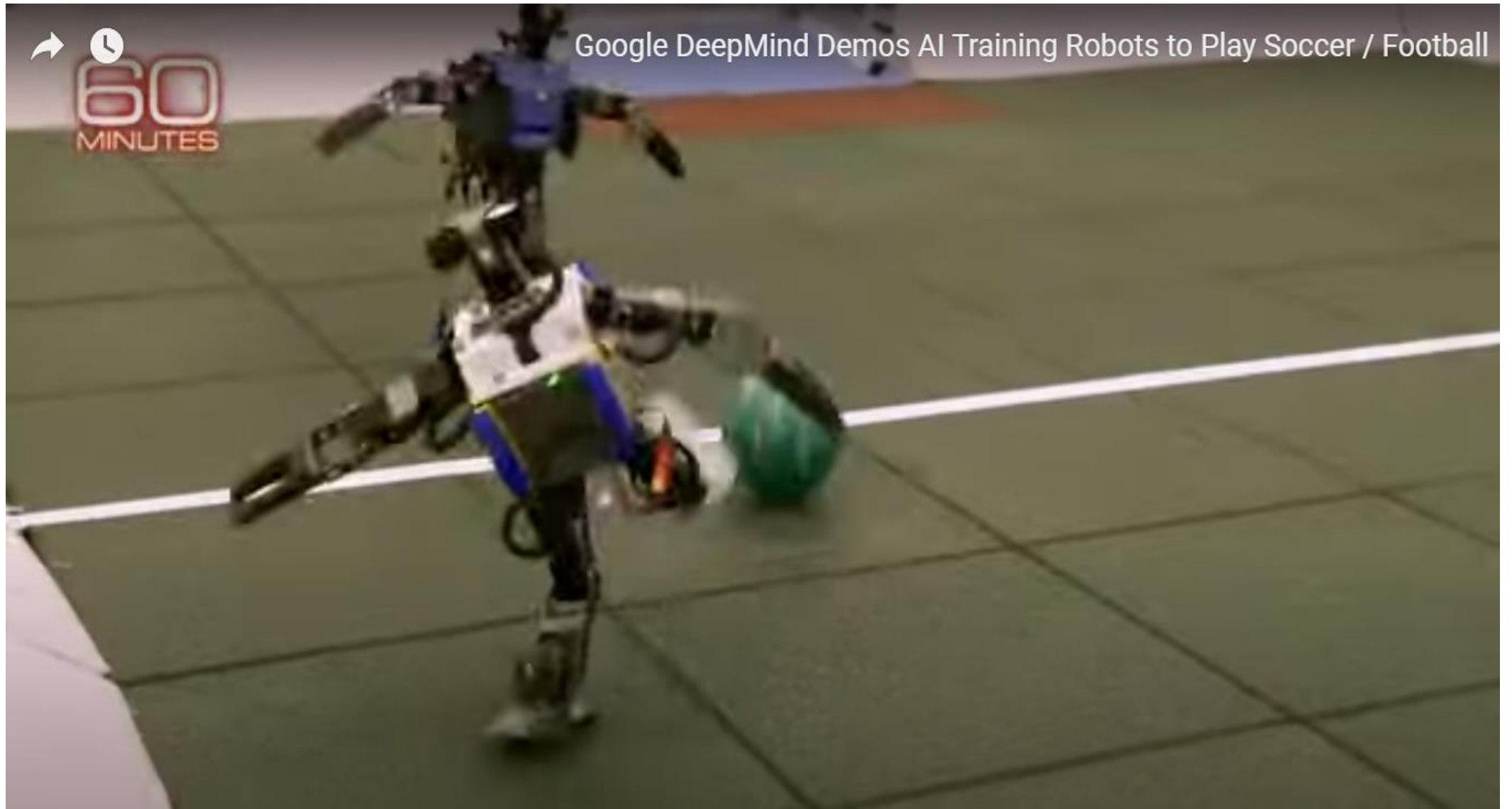
We're almost entering into science-fiction territory here, but ASI is seen as the logical progression from AGI. An Artificial Super Intelligence (ASI) system would be able to surpass all human capabilities. This would include decision making, taking rational decisions, and even includes things like making better art and building emotional relationships.

Once we achieve Artificial General Intelligence, AI systems would rapidly be able to improve their capabilities and advance into realms that we might not even have dreamed of. While the gap between AGI and ASI would be relatively narrow (some say as little as a nanosecond, because that's how fast Artificial Intelligence would learn) the long journey ahead of us towards AGI itself makes this seem like a concept that lays far into the future.

https://www.youtube.com/watch?v=W0_DPiOPmF0



<https://www.youtube.com/watch?v=RbyQcCT6890>



<https://www.youtube.com/shorts/o36sxpV2NRc>



https://twitter.com/Ronald_vanLoon/status/1207873508572684289/video/1



<https://www.youtube.com/watch?v=LOPbSHdWJ5E>



Difference between Human and Machine Intelligence

- Humans perceive by patterns whereas the machines perceive by set of rules and data.
- Humans store and recall information by patterns, machines do it by searching algorithms. For example, the number 40404040 is easy to remember, store and recall as its pattern is simple.
- Humans can figure out the complete object even if some part of it is missing or distorted; whereas the machines cannot correctly.

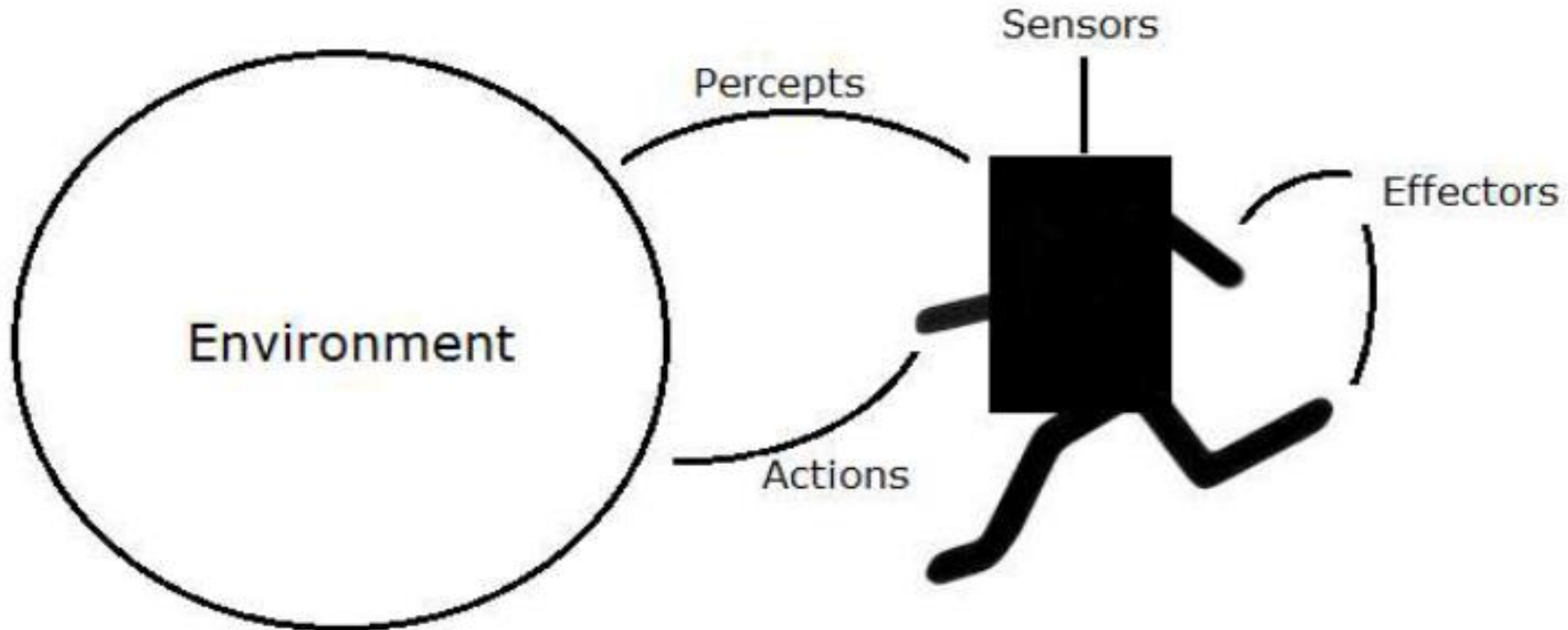
Speech and Voice Recognition

These both terms are common in robotics, expert systems and natural language processing. Though these terms are used interchangeably, their objectives are different.

Speech Recognition	Voice Recognition
The speech recognition aims at understanding and comprehending WHAT was spoken.	The objective of voice recognition is to recognize WHO is speaking.
It is used in hand-free computing, map or menu navigation	It analyzes person's tone, voice pitch, and accent, etc., to identify a person.
Machine does not need training as it is not speaker dependent.	The recognition system needs training as it is person-oriented.
Speaker independent Speech Recognition systems are difficult to develop.	Speaker-dependent Speech Recognition systems are comparatively easy to develop.

An agent

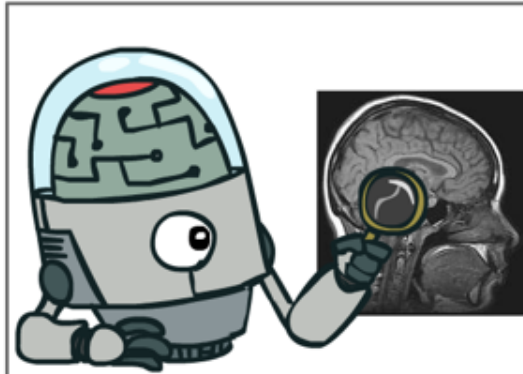
Is just something that acts. Of course, all computer programs do something, but computer agents are expected to do more: operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals.



What is AI?

The science of making machines that:

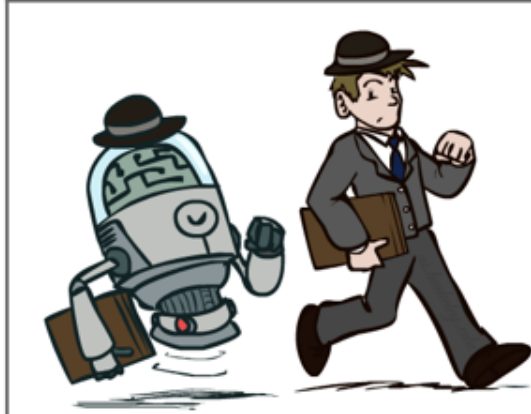
Think like people



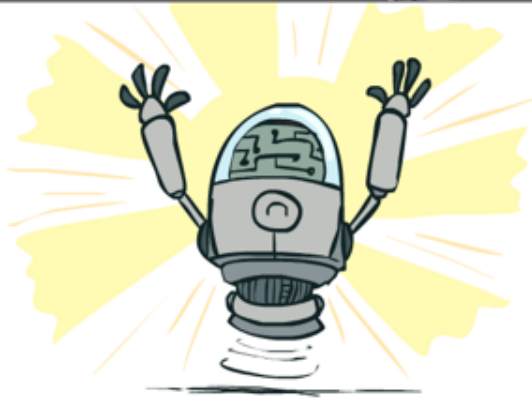
Think rationally



Act like people



Act rationally



WHAT IS AI?

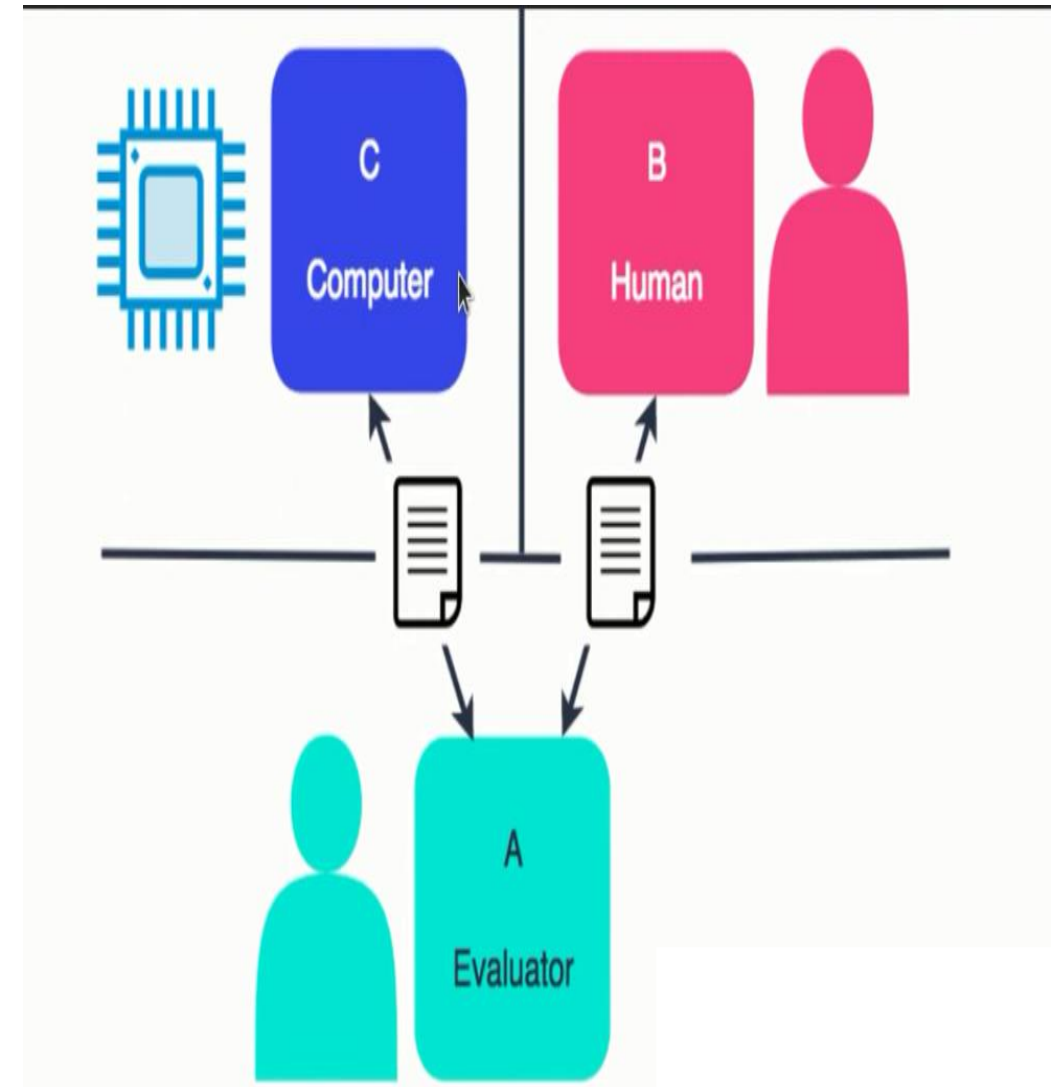
- The definitions on top are concerned with *thought processes* and *reasoning*, whereas the ones on the bottom address *behavior*.
- The definitions on the left measure success in terms of fidelity to *human* performance, whereas the ones on the right measure against an *ideal* performance measure, called **rationality**. A system is rational if it does the “right thing,” given what it knows.

Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i>, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i>, 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)
Figure 1.1 Some definitions of artificial intelligence, organized into four categories.	

Acting humanly: The Turing Test approach

The Turing Test, proposed by Alan Turing (1950), was designed provide a satisfactory operational definition of intelligence.

A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.



Acting humanly: The Turing Test approach

The computer would need to possess the following capabilities:

- **Natural language processing** to enable it to communicate successfully in English;
- **Knowledge representation** to store what it knows or hears;
- **Automated reasoning** to use the stored information to answer questions and to draw new conclusions;
- **Machine learning** to adapt to new circumstances and to detect and extrapolate patterns.

Thinking humanly: The cognitive modeling approach

If we are going to say that a given program thinks like a human, we must have some way of determining how humans think. We need to get *inside* the actual workings of human minds.

There are three ways to do this:

- Through introspection—trying to catch our own thoughts as they go by;
- Through psychological experiments—observing a person in action; and
- Through brain imaging—observing the brain in action.

Rational Decisions

We'll use the term **rational** in a very specific, technical way:

- Rational: maximally achieving pre-defined goals
- Rationality only concerns what decisions are made
(not the thought process behind them)
- Goals are expressed in terms of the **utility** of outcomes
- Being rational means **maximizing your expected utility**

A **rational agent** is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

Thinking rationally: The “laws of thought” approach

- Aristotle: what are correct arguments/thought processes?
- Several Greek schools developed various forms of *logic: notation* and *rules of derivation* for thoughts; may or may not have proceeded to the idea of mechanization
- Direct line through mathematics and philosophy to modern AI
- Problems:
 1. Not all intelligent behavior is mediated by logical deliberation
 2. What is the purpose of thinking? What thoughts should I have?

Acting rationally: The rational agent approach

Rational behavior: doing the right thing

The right thing: that which is expected to maximize goal achievement, given the available information

Doesn't necessarily involve thinking – e.g., blinking reflex – but thinking should be in the service of rational action

AI prehistory

Philosophy	Logic, methods of reasoning, mind as physical system foundations of learning, language, rationality
Mathematics	Formal representation and proof algorithms, computation, (un)decidability, (in)tractability, probability
Economics	utility, decision theory
Neuroscience	physical substrate for mental activity
Psychology	phenomena of perception and motor control, experimental techniques
Computer engineering	building fast computers
Control theory	design systems that maximize an objective function over time
Linguistics	knowledge representation, grammar

THE STATE OF THE ART

What can AI do today? A concise answer is difficult because there are so many activities in so many subfields.

- Robotic vehicles
- Speech recognition
- Autonomous planning and scheduling
- Game playing
- Spam fighting
- Logistics planning
- Robotics
- Machine Translation

Real Life Applications of Research Areas

There is a large array of applications where AI is serving common people in their day-to-day lives:

Sr. No.	Research Area	Real Life Application
1	Expert Systems Examples: Flight-tracking systems, Clinical systems	
2	Natural Language Processing Examples: Google Now feature, speech recognition, Automatic voice output	

3	Neural Networks Examples: Pattern recognition systems such as face recognition, character recognition, handwriting recognition.	
4	Robotics Examples: Industrial robots for moving, spraying, painting, precision checking, drilling, cleaning, coating, carving etc.	